

Test Plan Document

OpenCart Demo – Online Shopping Platform

Objectives

The goal is to ensure the platform's core e-commerce functionality (user registration, product browsing, cart management, checkout, and admin workflows) performs as expected and meets performance, usability, and security standards.

Approach

The test strategy combines manual and automated testing.

- Manual Testing: for usability, exploratory, and UI/UX validation.
- Automated Testing: for regression, API, and database validation.

Test Types: Functional, Integration, Performance, Security, and Usability testing.

Entry Criteria

Testing for the OpenCart Demo project will start only after the following conditions are met:

- The OpenCart application is deployed and accessible in the test environment.
- Core features (user registration, product browsing, cart, and checkout) are implemented and stable.
- Test data (sample users, products, and admin credentials) is prepared.
- All required APIs and databases are connected and functional.
- Test cases are written, reviewed, and approved by the QA Lead.
- The testing environment (browsers, devices, network) is ready and verified.

Exit Criteria

Testing will be considered complete when the following conditions are satisfied:

- All planned test cases (manual and automated) have been executed.
- All critical and major defects have been fixed, verified, and closed.
- Regression testing has been completed successfully.
- No high-severity issues remain open that block core functionality.
- The final test summary report has been prepared and approved by the QA Lead.
- All test artifacts (test cases, defect reports, execution logs) are documented and stored for future reference.

Scope

In Scope:

- User registration and login
- Product browsing and search functionality
- Cart management (add, update, remove items)
- Checkout process and payment gateway testing

- Admin workflows (product CRUD, user management)
- API and database validation

Out of Scope:

- Third-party plugin customization
- Mobile native app (web testing only)
- Non-production payment gateways

Test Deliverables

- Test cases (manual and automated)
- Defect reports and test execution reports
- Final test summary report

Deliverables will be shared bi-weekly with stakeholders and archived for future regression testing.

Dependencies

- Database setup (MySQL)
- Admin credentials for backend access
- Payment gateway sandbox accounts (e.g., PayPal, Stripe)
- Stable internet connection and server availability

Contingency plan: Use mock data and sandbox APIs if live integrations are unavailable.

Test Environment

- Devices: Desktop (Windows 10/11, macOS), Mobile browsers (Android/iOS)
- Browsers: Chrome, Firefox, Edge, Safari
- Server: Linux with MySQL
- Network: Wi-Fi, 4G

Risk Management

Identified Risks:

- Third-party payment API downtime
- Database connection errors
- Performance degradation under high traffic

Mitigation Strategies:

- Use mock payment data for failed integrations
- Conduct load testing to identify performance bottlenecks
- Regular DB backup before tests

Schedule

Timeline:

- Test Planning: Oct 17 – Oct 18

- Test Case Design: Oct 19 – Oct 21
- Functional Testing: Oct 22 – Oct 25
- Performance & Security Testing: Oct 26 – Oct 27
- Test Reporting: Oct 28

Roles and Responsibilities

- QA Lead: oversees test planning, execution, and reporting
- Test Engineers: executes test cases, logs defects
- Developers: fixes bugs during testing
- DevOps: maintains the test environment